Registration form for Minor Project-I

Roll No	Regd No	Name of the Student
23CSE485	23UG010568	DEEPAK PANDA
23CSE442	23UG010525	PRANJAL PRAKASH
23CSE400	23UG010483	RAHUL BISOYI

Proposed Project Title:D-SPAN

Supervisor Name:mr. Ranjit Pattnaik

Abstract:

The DNA Sequence Pattern Analyzer (D-SPAN) is an interactive web-based application designed to simplify and visualize fundamental bioinformatics operations on DNA sequences. Developed using front-end technologies such as HTML, CSS, and JavaScript, D-SPAN provides an accessible platform for students, educators, and researchers to explore the structure and features of DNA in a hands-on way. The tool allows users to input DNA sequences and perform a variety of biologically meaningful analyses, including generating the complementary strand, calculating the GC content, finding the reverse complement, searching for custom patterns, and detecting microsatellite repeats (short tandem repeats like AGAGAG or TATATA). These features are vital for understanding core genetic processes such as DNA replication, transcription, mutation detection, gene regulation, and DNA fingerprinting.

One of the key educational values of D-SPAN is its ability to simulate real biological mechanisms in a digital environment. For instance, the "Generate Complement" feature mimics how complementary base pairing occurs during replication, while the "GC Content" calculation highlights the relationship between sequence composition and DNA stability. The "Custom Pattern Search" helps identify known motifs or potential mutations, and the "Microsatellite Detection" feature is particularly relevant in studying genetic variation and identity, as these repeats are often used as genetic markers in forensics and population genetics.

The tool also introduces users to the importance of sequence orientation through reverse complement analysis, which reflects the way genetic information is read in living organisms. By automating these analyses in a browser-based interface, D-SPAN eliminates the need for advanced bioinformatics software and makes genetic pattern exploration more accessible.

Official Use		
Group No:	**Section:G**	**Category:** Bioinformatics Web Application ☐

Supervisor **Class Teacher** **Mr.Bhavani Sankar Panda**
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